HARDWARE LABORATORY 3

TECHNOLOGIES FOR IOT ECOSYSTEMS

Prof. Luca Barbierato

Assistant Professor

Politecnico di Torino

1859

Electronic Design Automation Group

Dipartimento di Automatica
ed Informatica

Overview

All three exercises in Part 3 use the same HW components

- One of the two LEDs (or the internal LED)
- The temperature sensor (or the internal temperature sensor)
- The Serial interface (for debugging and error reporting)
- The WiFi interface of the Arduino

The goal is to let the Arduino communicate via REST and MQTT

luca.barbierato@polito.it

EXERCISE 1

TECHNOLOGIES FOR IOT ECOSYSTEMS

Prof. Luca Barbierato

Assistant Professor

Politecnico di Torino

1859

Electronic Design Automation Group

Dipartimento di Automatica ed Informatica

The WiFiNINA Library

Used to control the WiFi and BLE Interface of the RP2040.

- Reference: here

In this exercise we will use two components of the library

- `WiFiServer` to setup an HTTP server on the Arduino
- `WiFiClient` to handle individual requests to the server

New Code Elements: Global

Include required libraries and define the SSID (aka “network name”) and Password of the WiFi network you want to connect to.

```
#include <WiFiNINA.h>
#include "arduino_secrets.h"

char ssid[] = SECRET_SSID;
char pass[] = SECRET_PASS;
```

In the lab, you must use your phone’s hotspot. The “Polito” WiFi won’t work.

The best practice suggested by Arduino is to define your SSID and Password in a different header file (“arduino_secrets.h”)

Better than defining them in the .ino “main file”, because it’s more difficult to accidentally share “secrets”.

```
#define SECRET_SSID "XXXXXXXXXX"
#define SECRET_PASS "XXXXXXXXXX"
```

Careful, this is not really secure!!! Delete this file after testing, and **don’t send me your WiFi passwords when you upload the labs!**

New Code Elements: Global

Define a global variable to hold the WiFi connection status.

```
int status = WL_IDLE_STATUS;
```

Define a `WiFiServer` object that will listen on Port 80 (HTTP)

```
WiFiServer server(80);
```

New Code Elements: Setup

In `setup()`:

1. Try connecting to the WiFi until you succeed.

2. When connected, print the IP address assigned by DHCP on the Serial.

   You need to know the IP address in order to connect to the Arduino from your PC.

3. Start the server on Port 80

```
void setup() {

  //other code...

  while (status != WL_CONNECTED) {
    Serial.print("Attempting to connect to SSID: ")
    Serial.println(ssid);
    status = WiFi.begin(ssid, pass);
    delay(10000);
  }

  Serial.print("Connected with IP Address: ");
  Serial.println(WiFi.localIP());

  server.begin();

}
```

New Code Elements: Loop

In `loop()`:

1. Check if a connection (client) is available

2. Process the incoming connection and terminate it

3. Wait for 50ms to avoid overloading

```
void loop() {
  WiFiClient client = server.available();
  if (client) {
    process(client);
    client.stop();
  }
  delay(50);
}
```

New Code Elements: Process

You can read from the client as any other `Stream`, exactly as the `Serial` class

The HTTP request “start line” tells you the type of request and the URL of the requested resource. Example:

```
GET /led/1
```

Parse this line:

```
void process(WiFiClient client) {

  String req_type = client.readStringUntil(' ');
  req_type.trim();
  String url = client.readStringUntil(' ');
  url.trim();
```

Read until next space

Delete \n or similar

New Code Elements: Process

Respond based on the request type and URL:

Note: You should implement more robust checks than these…

```
if (url.startsWith("/led/")) {
  String led_val = url.substring(5);
  Serial.println(led_val);
  if (led_val == "0" || led_val == "1") {
    int int_val = led_val.toInt();
    digitalWrite(LED_PIN, int_val);
    printResponse(client, 200, senMlEncode("led", int_val, ""));
  } else {
    // etc...
  }
```

New Code Elements: printResponse

In the `printResponse()` function, print the header and body of the HTTP response

HTTP response start line

```
void printResponse(WiFiClient client, int code, String body) {
  client.println("HTTP/1.1 " + String(code));
  if (code == 200) {
    client.println("Content-type: application/json; charset=utf-8");
    client.println(); //mandatory blank line
    client.println(body); //the response body
  } else{
    client.println();
  }
}
```

Here the `Content-type` header helps having a better formatting in the browser (in the next exercise it will be fundamental)

New Code Elements

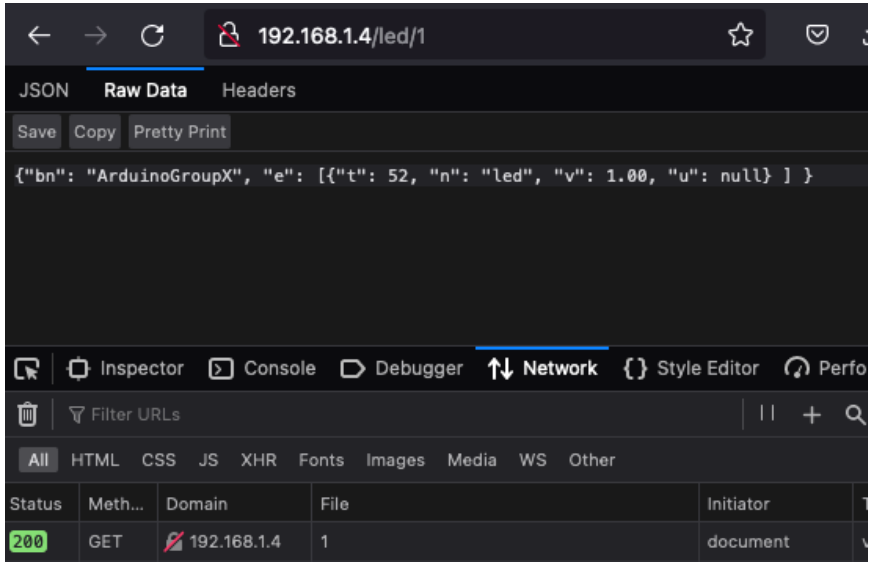
The body must be encoded in SenML JSON format (job of the `senMlEncode` function in my example):

```
{
  "bn": "ArduinoGroupX"
  "e": [
      {
      "n": <"temperature">/<"led">,
      "t": <timestamp using millis()>,
      "v": value,
      "u": "Cel"/null
      }
  ]
}
```

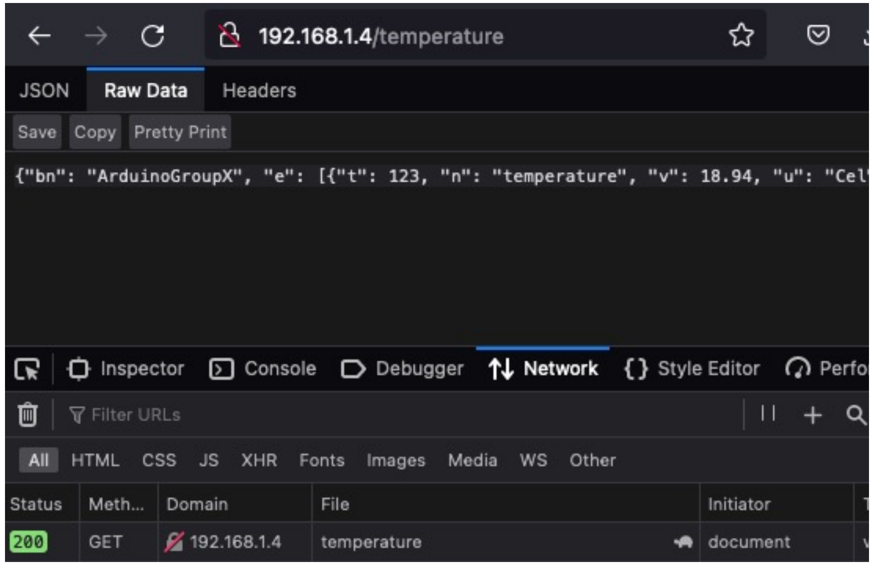
You can do this "by hand" (it's just a concatenation of strings):

- In Exercise 3, we'll see how to do more flexible encoding and (most importantly) decoding with the `ArduinoJson` library.

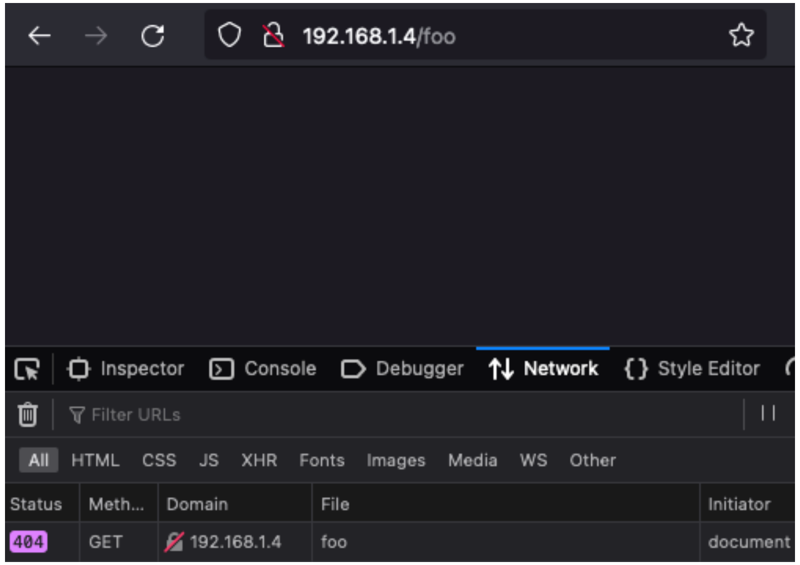
Results



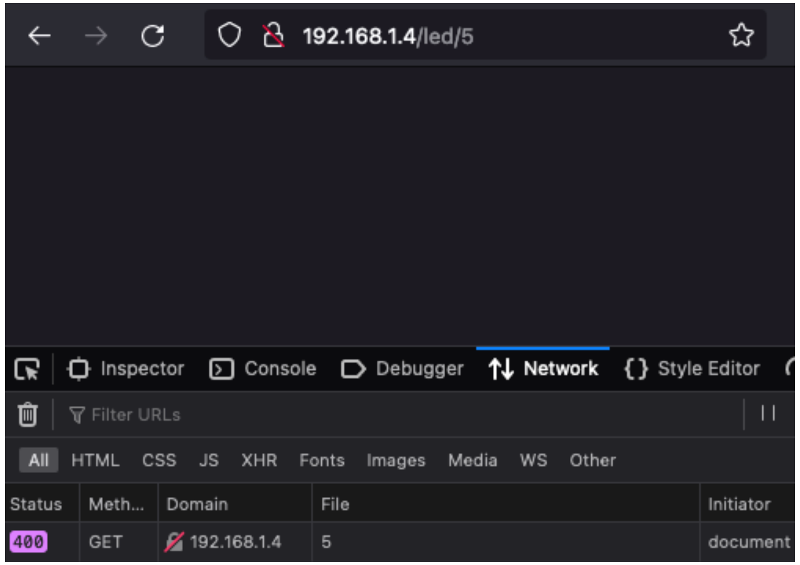
192.168.1.4/led/1
JSON
Raw Data
Headers
Save
Copy
Pretty Print
{"bn": "ArduinoGroupX", "e": [{"t": 52, "n": "led", "v": 1.00, "u": null}] }
Inspector
Console
Debugger
Network
Style Editor
Filter URLs
All HTML CSS JS XHR Fonts Images Media WS Other
Status
Meth...
Domain
File
Initiator
200
GET
192.168.1.4
1
document



192.168.1.4/temperature
JSON
Raw Data
Headers
Save
Copy
Pretty Print
{"bn": "ArduinoGroupX", "e": [{"t": 123, "n": "temperature", "v": 18.94, "u": "Cel
Inspector
Console
Debugger
Network
Style Editor
Filter URLs
All HTML CSS JS XHR Fonts Images Media WS Other
Status
Meth...
Domain
File
Initiator
200
GET
192.168.1.4
temperature
document



192.168.1.4/foo
Inspector
Console
Debugger
Network
Style Editor
Filter URLs
All HTML CSS JS XHR Fonts Images Media WS Other
Status
Meth...
Domain
File
Initiator
404
GET
192.168.1.4
foo
document



192.168.1.4/led/5
Inspector
Console
Debugger
Network
Style Editor
Filter URLs
All HTML CSS JS XHR Fonts Images Media WS Other
Status
Meth...
Domain
File
Initiator
400
GET
192.168.1.4
5
document

EXERCISE 2

TECHNOLOGIES FOR IOT ECOSYSTEMS

Prof. Luca Barbierato

Assistant Professor

Politecnico di Torino

1859

Electronic Design Automation Group

Dipartimento di Automatica ed Informatica

Arduino HTTP Client

Include libraries:

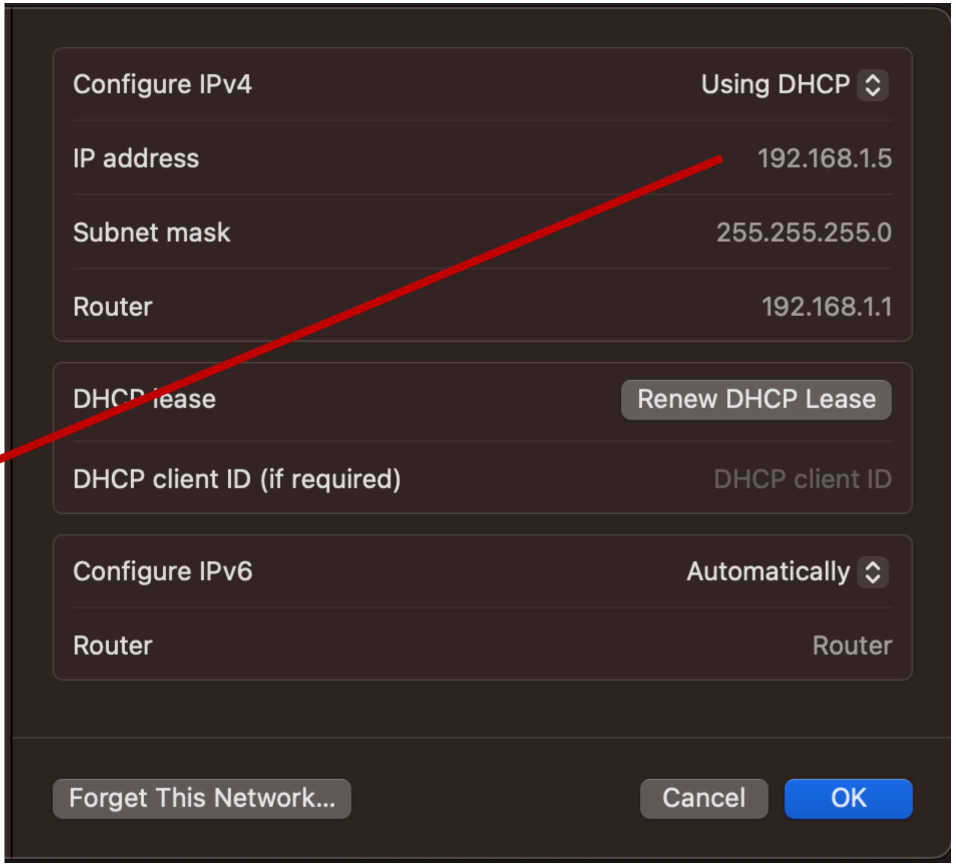
```
#include <WiFiNINA.h>
#include <ArduinoHttpClient.h>
#include "arduino_secrets.h"
```

- `ArduinoHTTPClient` reference (link)

Define the server address and port:

```
char server_address[] = "192.168.1.5";
int server_port = 8080;
```

How to get your computer IP depends on the OS:



Arduino HTTP Client

Declare a `WiFiClient` instance directly

- Not obtained from a server as in the previous exercise

Declare a `HttpClient` instance, passing the `WiFiClient`, IP address and port

- HttpClient facilitates the creation of HTTP requests
- …so that you don't have to deal with the protocol at low level (managing basic strings)

```
WiFiClient wifi;
HttpClient client = HttpClient(wifi, server_address, server_port);
```

Arduino HTTP Client

The setup() function is identical to the previous exercise.

In the loop():

```
void loop() {
  // read temperature and create a "body" String in SenML
  // format...

  client.beginRequest();
  client.post("/log");
  client.sendHeader("Content-Type", "application/json");
  client.sendHeader("Content-Length", body.length());
  client.beginBody();
  client.print(body);
  client.endRequest();
  int ret = client.responseStatusCode();
```

Type of request and URL

Headers. Both content type and length are very important

Add the body

Get the return code.

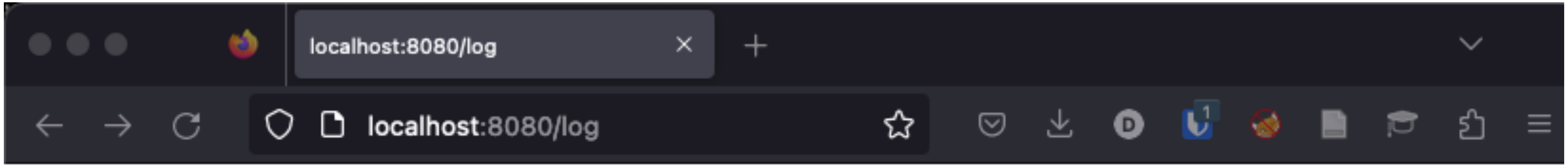
Python HTTP Server

You also have to **modify the cherrypy servers** developed in Exercises 1 and 2 of the SW lab to handle both POST and GET requests to the resource:

`http://<PC IP Address>:<port>/log`

Nothing special about this, you just need to **store all logged data** and do some JSON `loads()` and `dumps()`....

Server GET Results



[{"bn": "ArduinoGroupX", "e": [{"t": 19, "n": "temperature", "v": 18.78238487, "u": "Cel"}]}, {"bn": "ArduinoGroupX", "e": [{"t": 21, "n": "temperature", "v": 18.78238487, "u": "Cel"}]}, {"bn": "ArduinoGroupX", "e": [{"t": 23, "n": "temperature", "v": 18.78238487, "u": "Cel"}]}, {"bn": "ArduinoGroupX", "e": [{"t": 26, "n": "temperature", "v": 18.78238487, "u": "Cel"}]}, {"bn": "ArduinoGroupX", "e": [{"t": 28, "n": "temperature", "v": 18.78238487, "u": "Cel"}]}, {"bn": "ArduinoGroupX", "e": [{"t": 30, "n": "temperature", "v": 18.78238487, "u": "Cel"}]}, {"bn": "ArduinoGroupX", "e": [{"t": 34, "n": "temperature", "v": 18.70260048, "u": "Cel"}]}, {"bn": "ArduinoGroupX", "e": [{"t": 36, "n": "temperature", "v": 18.70260048, "u": "Cel"}]}, {"bn": "ArduinoGroupX", "e": [{"t": 38, "n": "temperature", "v": 18.86216354, "u": "Cel"}]}, {"bn": "ArduinoGroupX", "e": [{"t": 42, "n": "temperature", "v": 18.78238487, "u": "Cel"}]}, {"bn": "ArduinoGroupX", "e": [{"t": 44, "n": "temperature", "v": 18.86216354, "u": "Cel"}]}, {"bn": "ArduinoGroupX", "e": [{"t": 48, "n": "temperature", "v": 18.78238487, "u": "Cel"}]}, {"bn": "ArduinoGroupX", "e": [{"t": 50, "n": "temperature", "v": 18.70260048, "u": "Cel"}]}]

EXERCISE 3

TECHNOLOGIES FOR IOT ECOSYSTEMS

Prof. Luca Barbierato

Assistant Professor

Politecnico di Torino
1859

Electronic Design Automation Group
Dipartimento di Automatica ed Informatica

Arduino MQTT: Global

Declare the URL (or IP address) of the broker and the prefix of the topic:

```
String broker_address = "test.mosquitto.org";
int broker_port = 1883;

const String base_topic = "/tiot/0";
```


luca.barbierato@polito.it

Arduino MQTT: Global

Create two global objects to store the sent and received JSON strings:

```
// Enough space for 1 SenML record (plus spare)
const int capacity = JSON_OBJECT_SIZE(2) + JSON_ARRAY_SIZE(1) + JSON_OBJECT_SIZE(4) + 100;
DynamicJsonDocument doc_snd(capacity);
DynamicJsonDocument doc_rec(capacity);
```


luca.barbierato@polito.it

Arduino MQTT: Callback

Define a callback to handle messages arriving on subscribed topics:

- ArduinoJson helps you transform the received string (array of bytes) into an easily-accessible object:

"topic" can be used to share a single callback for multiple topics

```
void callback(char* topic, byte* payload, unsigned int length) {
  DeserializationError err = deserializeJson(doc_rec, (char*) payload);
  if (err) {
    Serial.print(F("deserializeJson() failed with code "));
    Serial.println(err.c_str());
  }
  if (doc_rec["e"][0]["n"] == "led") {
    if(doc_rec["e"][0]["v"] == 1) {
      //etc
```

Cast byte array into char array (C string)
Then de-serialize it, saving it into the `doc_rec` object.

.c_str() converts other string-like classes into C strings (arrays of chars)

Access the object's fields by name (key) or by index, based on the received JSON format (SenML in our case).

ArduinoJson Details

With `ArduinoJson` you can also check if:

- The JSON was correctly parsed: `.isNull()` method
- A given field is present in the received object: `.containsKey()` method

Moreover, you can also use it as a better alternative to manual String concatenation to generate JSON strings (aka “serialize” a JSON object):

```
String senMlEncode(String res, float v, String unit) {
  doc_snd.clear();
  doc_snd["bn"] = "ArduinoGroup0";
  doc_snd["e"][0]["t"] = int(millis()/1000);
  // etc...
  String output;
  serializeJson(doc_snd, output);
  return output;
}
```

Initialized object

Add fields as needed

Serialize into a String

Arduino MQTT: Global

In the global section, declare a `PubSubClient` object:

- This must be done <u>after</u> defining the callback!
- Similarly to `ArduinoHTTPClient`, this object uses a `WiFiClient` internally to transmit/receive messages
- Other parameters are the broker's address and port, and the callback function just defined to handle incoming messages

```
WiFiClient wifi;
PubSubClient client(broker_address.c_str(), broker_port, callback, wifi);
```

Differently from ArduinoHTTPClient, the address must be a C char array.

Arduino MQTT: Loop

The `setup()` is identical to the previous 2 exercises

In the `loop()`:

```
void loop() {

  if(client.state() != MQTT_CONNECTED) {
    reconnect();
  }
  //read sensor and create json message body...

  client.publish((base_topic + String("/temperature")).c_str(), body.c_str());
  client.loop();
```

Publish a message: `publish(topic, body)`

Check if there are new messages on subscribed topics

Arduino MQTT: Reconnect

Function to connect (or re-connect) to the broker and subscribe (re-subscribe) to topics. Called by the loop():

```
void reconnect() {
  // Loop until connected
  while (client.state() != MQTT_CONNECTED) {
    if (client.connect("TiotGroup0")) {
      client.subscribe((base_topic + String("/led")).c_str());
    } else {
      Serial.print("failed, rc=");
      Serial.print(client.state());
      Serial.println(" try again in 5 seconds");
      delay(5000);
    }
  }
}
```

Connection requires a unique ClientID. Use your group number

`.subscribe(topic)`

Testing

To test the exercise, you'll need to install the Mosquitto client (or another MQTT client) on your laptop (instructions can be found here)

Subscribe command on your terminal (to read temperature logs):

```
mosquitto_sub -h test.mosquitto.org -t '/tiot/group0/temperature'
```

Publish command on your terminal (to turn ON the LED):

```
mosquitto_pub -h test.mosquitto.org -t '/tiot/group0/led' -m
'{"bn": "ArduinoGroup0", "e": [{"n": "led", "t": null, "v": 1,
"u": null}]}'
```

Results

```
➜ lab_3.2 mosquitto_sub -h test.mosquitto.org -t '/tiot/0/temperature'
{"bn":"ArduinoGroup0","e":[{"t":81,"n":"temperature","v":18.30358887,"u":"Cel"}]}
{"bn":"ArduinoGroup0","e":[{"t":86,"n":"temperature","v":18.30358887,"u":"Cel"}]}
{"bn":"ArduinoGroup0","e":[{"t":91,"n":"temperature","v":18.22376442,"u":"Cel"}]}
{"bn":"ArduinoGroup0","e":[{"t":96,"n":"temperature","v":18.46321487,"u":"Cel"}]}
{"bn":"ArduinoGroup0","e":[{"t":101,"n":"temperature","v":18.46321487,"u":"Cel"}]}
{"bn":"ArduinoGroup0","e":[{"t":106,"n":"temperature","v":18.30358887,"u":"Cel"}]}
{"bn":"ArduinoGroup0","e":[{"t":111,"n":"temperature","v":18.30358887,"u":"Cel"}]}
{"bn":"ArduinoGroup0","e":[{"t":116,"n":"temperature","v":18.46321487,"u":"Cel"}]}
{"bn":"ArduinoGroup0","e":[{"t":121,"n":"temperature","v":18.30358887,"u":"Cel"}]}
```

```
➜ ~ mosquitto_pub -h test.mosquitto.org -t '/tiot/0/led' -m '{"bn": "ArduinoGroup0", "e": [{"n": "led", "t": null, "v": 1, "u": null}]}'
➜ ~ mosquitto_pub -h test.mosquitto.org -t '/tiot/0/led' -m '{"bn": "ArduinoGroup0", "e": [{"n": "led", "t": null, "v": 0, "u": null}]}'
```

Prof. Luca Barbierato

Assistant Professor

Politecnico di Torino
1859

Electronic Design Automation Group
Dipartimento di Automatica
ed Informatica